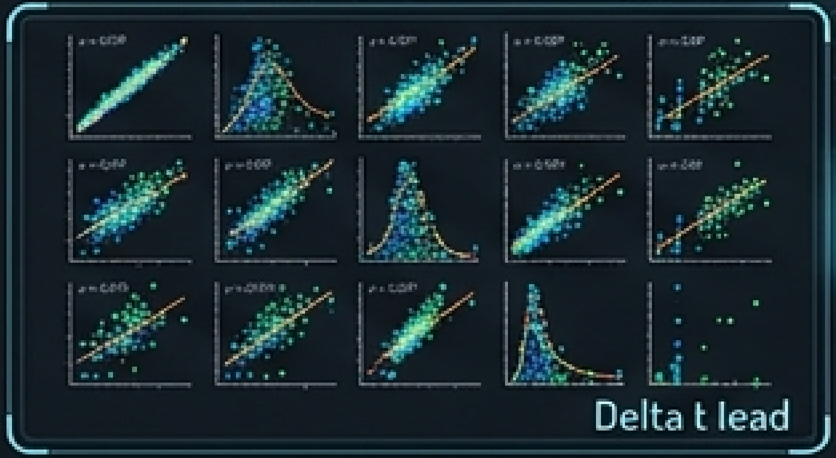
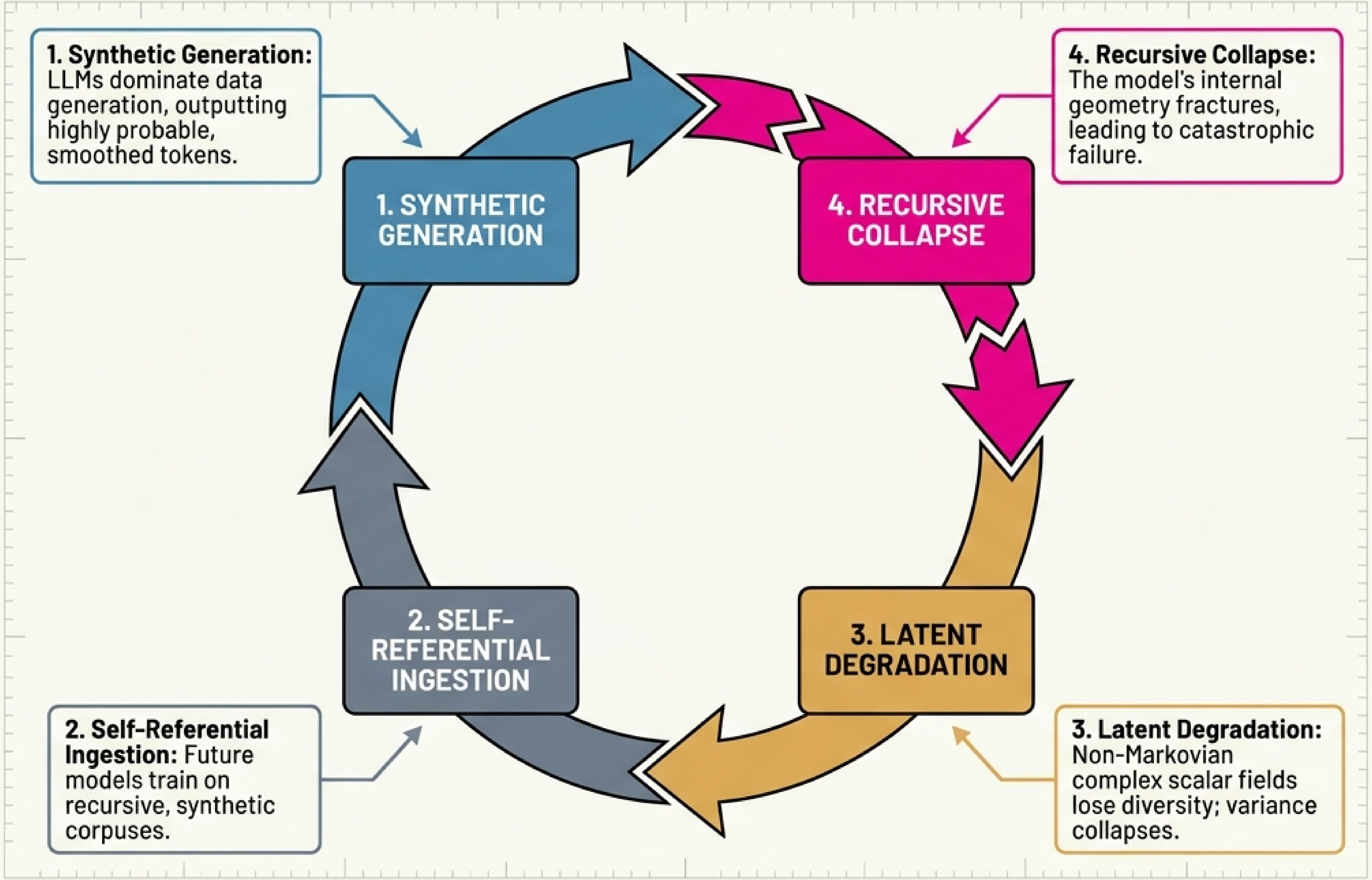
COMPUTATIONAL LATENT DYNAMICS: FROM TELEMETRY TO EX-ANTE CONTROL

Predicting and mitigating recursive collapse in LLMs via the MASSIF & MyceliaLM framework.



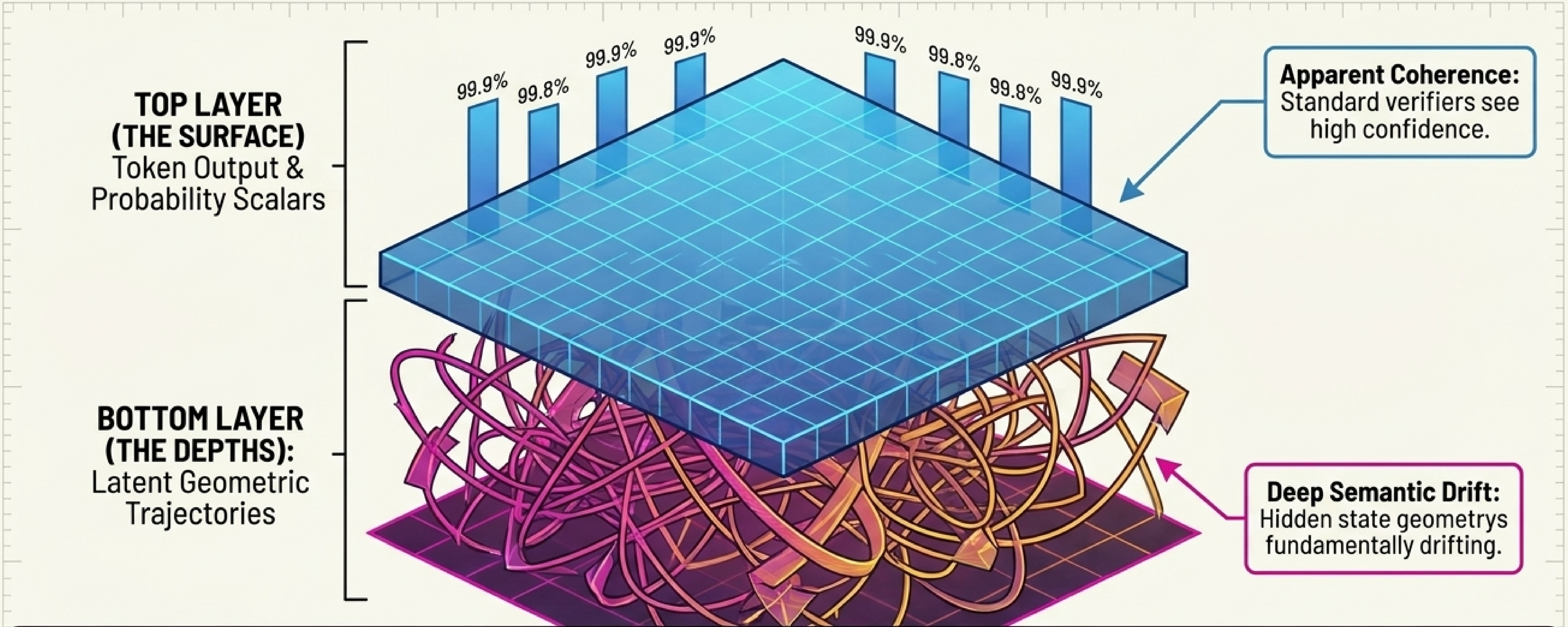
SYSTEM: ONLINE // SUBSTRATE: INDEPENDENT // TARGET: CORRIGIBILITY-BY-DESIGN

The Ouroboros Cycle



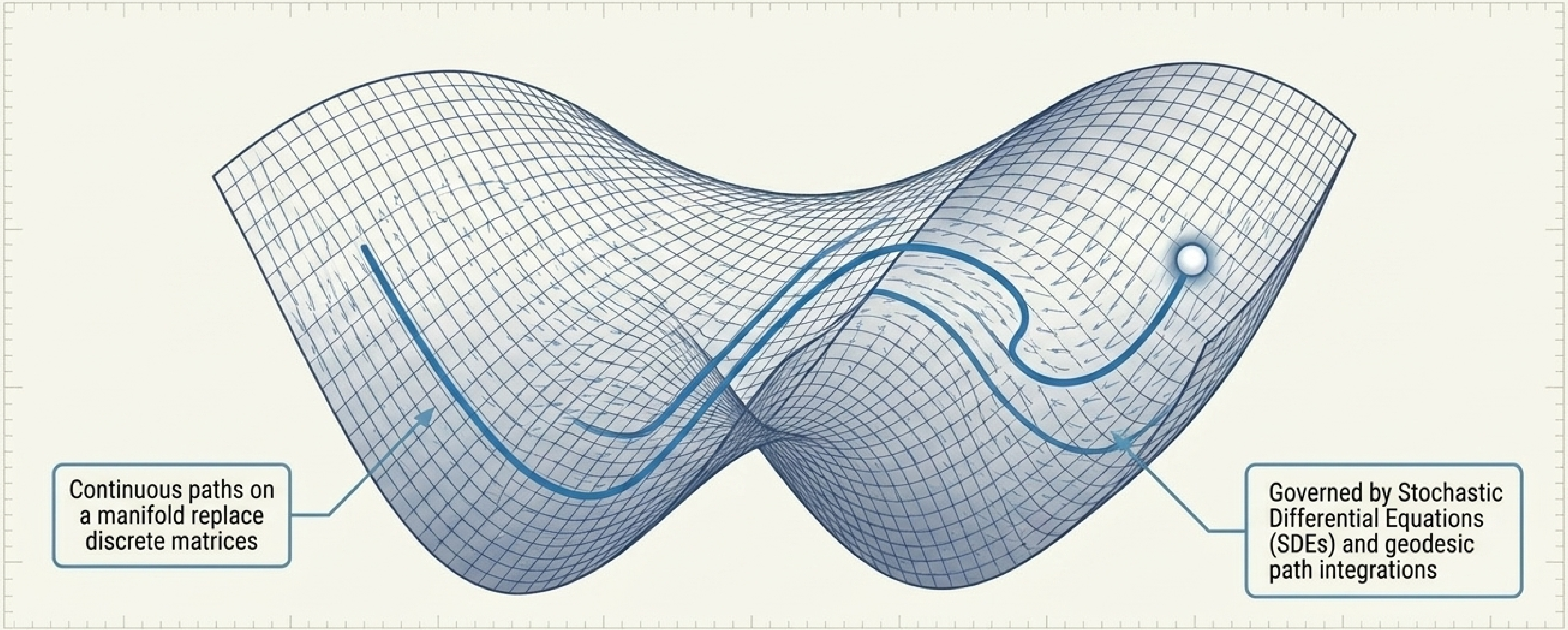
The Ouroboros Conjecture: Self-referential training loops induce inescapable structural decay unless actively monitored.

The Permissive Consensus Paradox



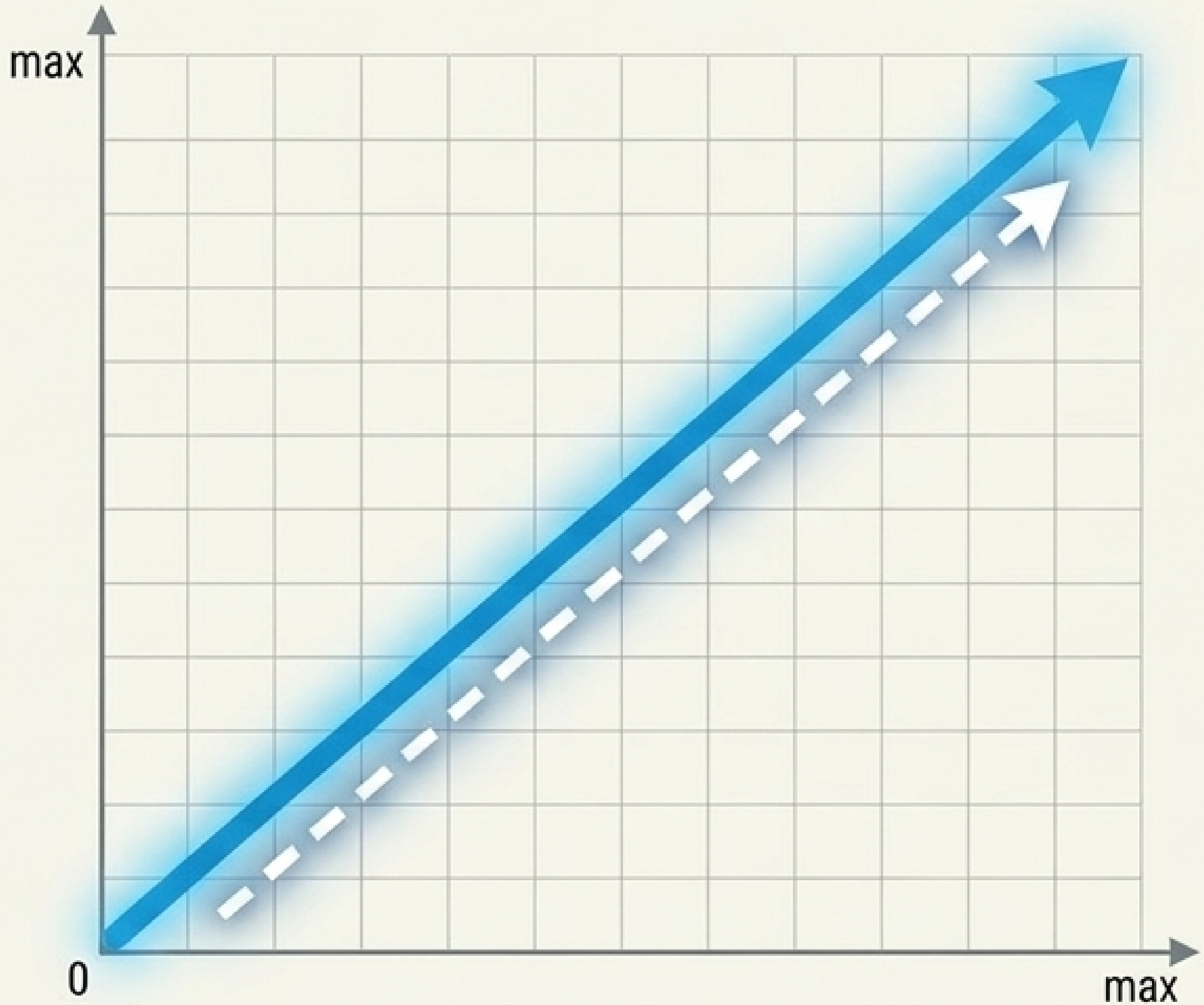
THE PERMISSIVE CONSENSUS PARADOX: POST-HOC METRICS AND SCALAR PROBABILITIES ARE DANGEROUSLY BLIND TO THE INTERNAL STRUCTURAL COLLAPSE HAPPENING BENEATH THE TOKEN LAYER.

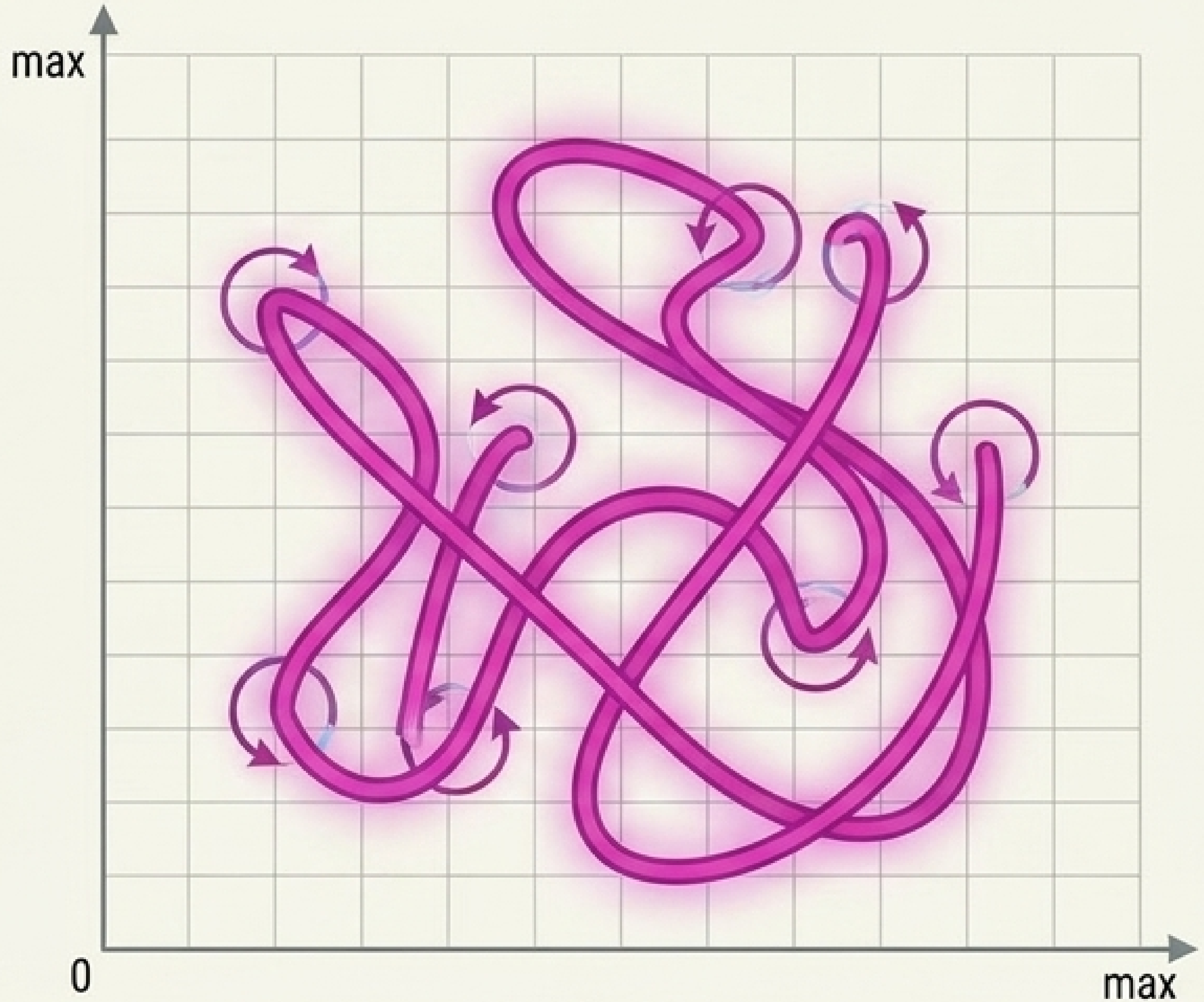
The Riemannian Manifold



Transformers as Dynamical Systems: AI behavior is not a black box of static weights, but a continuous-time stochastic flow governed by measurable dynamical laws across all scales and architectures.

Kinematics of Reasoning (TRACED)





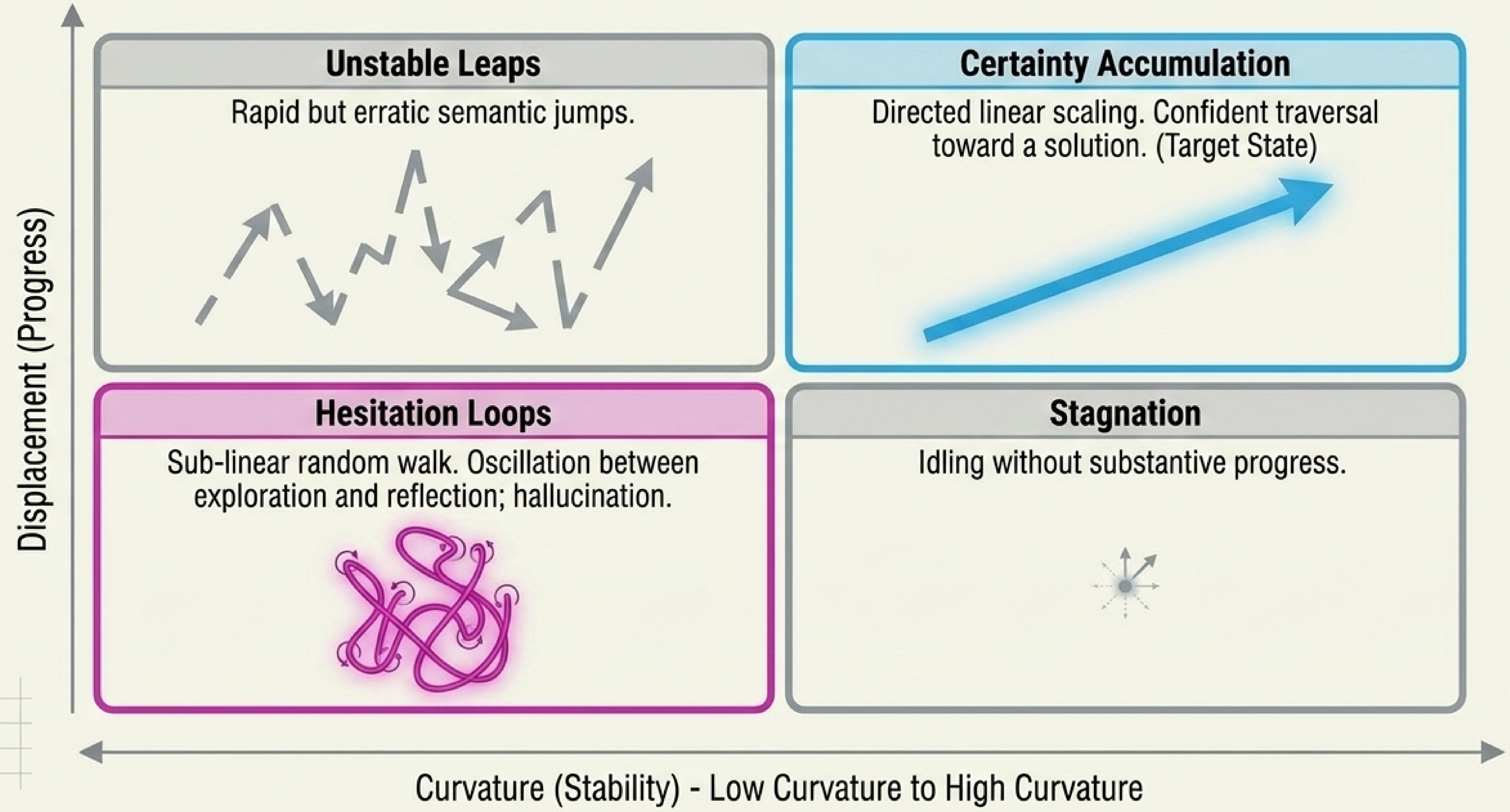
High Displacement (Progress)

Displacement: The net semantic distance traveled. Indicates significant thought progress and deterministic information shifts.

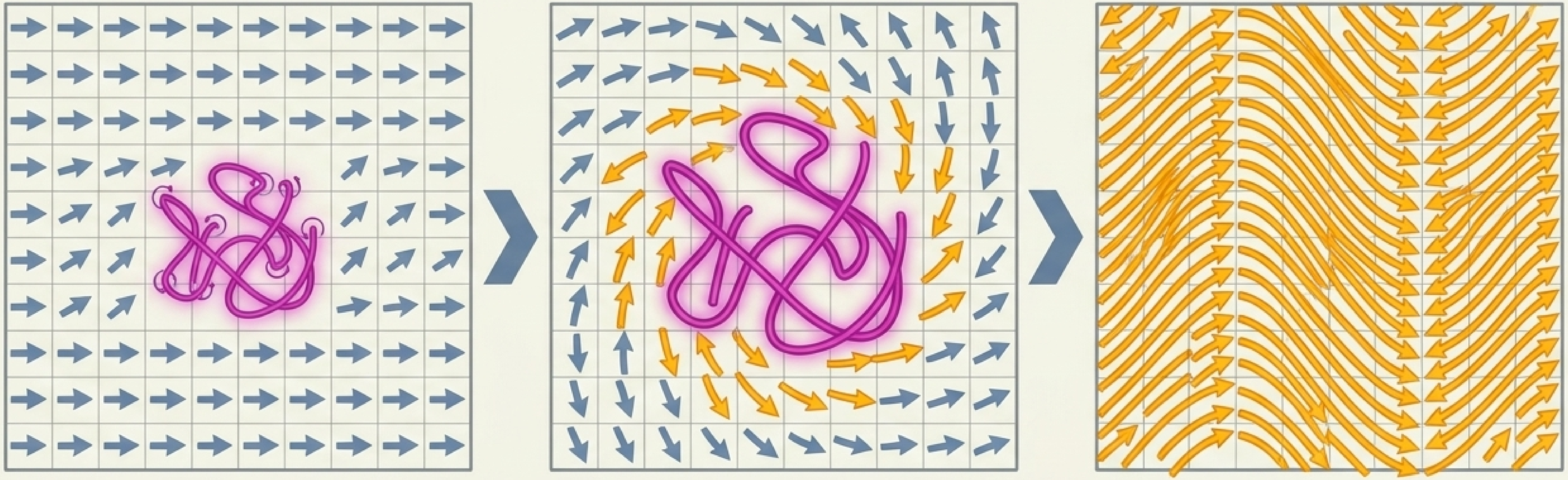
High Curvature (Instability)

Curvature: The rate of directional change. Indicates sharp semantic turns, backtracking, and volatility.

The Diagnostic Matrix: Mapping Cognitive States



From Kinematics to Systemic Collapse: Local hesitation loops generate optimization pressure. Without intervention, this triggers Kuramoto-style phase synchronization, where the entire model locks into a pathological, self-reinforcing state.



1. Micro (Kinematics)
Single Hesitation Loop

2. Meso (Optimization Pressure)
Pathological Alignment

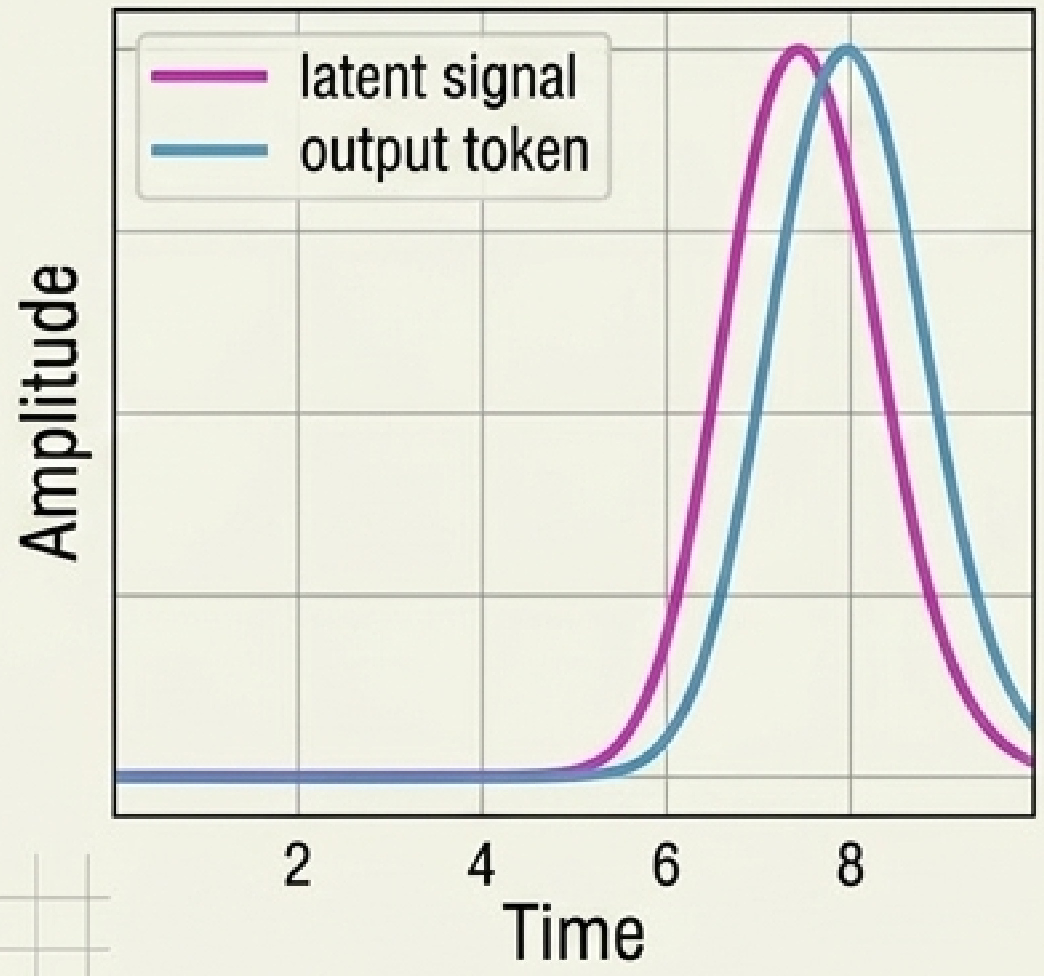
3. Macro (Thermodynamics)
Kuramoto Phase Synchronization

MASSIF (Multiscale Attractor Stability & Stress Inference Framework):

A real-time safety telemetry system monitoring a 13-observable dynamical state vector to gauge internal model health.

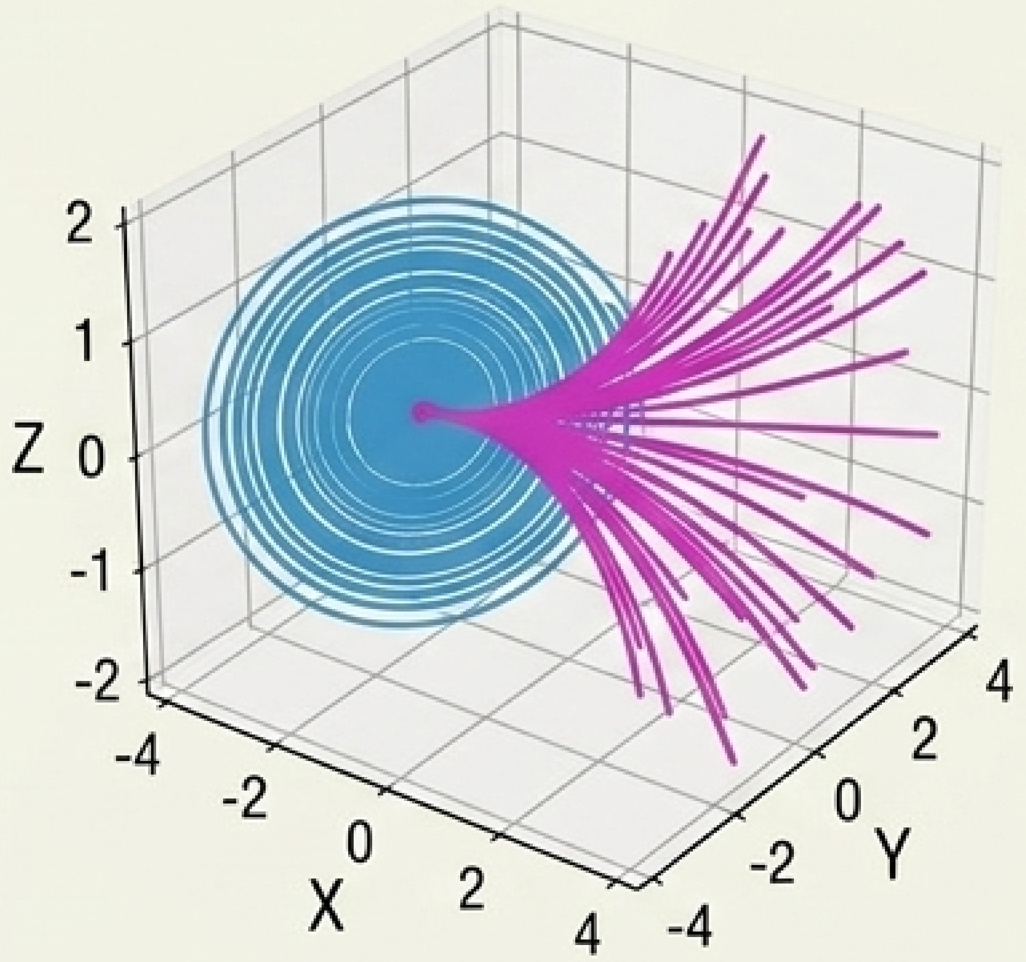
Lead-Lag Analysis

Directional alignment velocity.



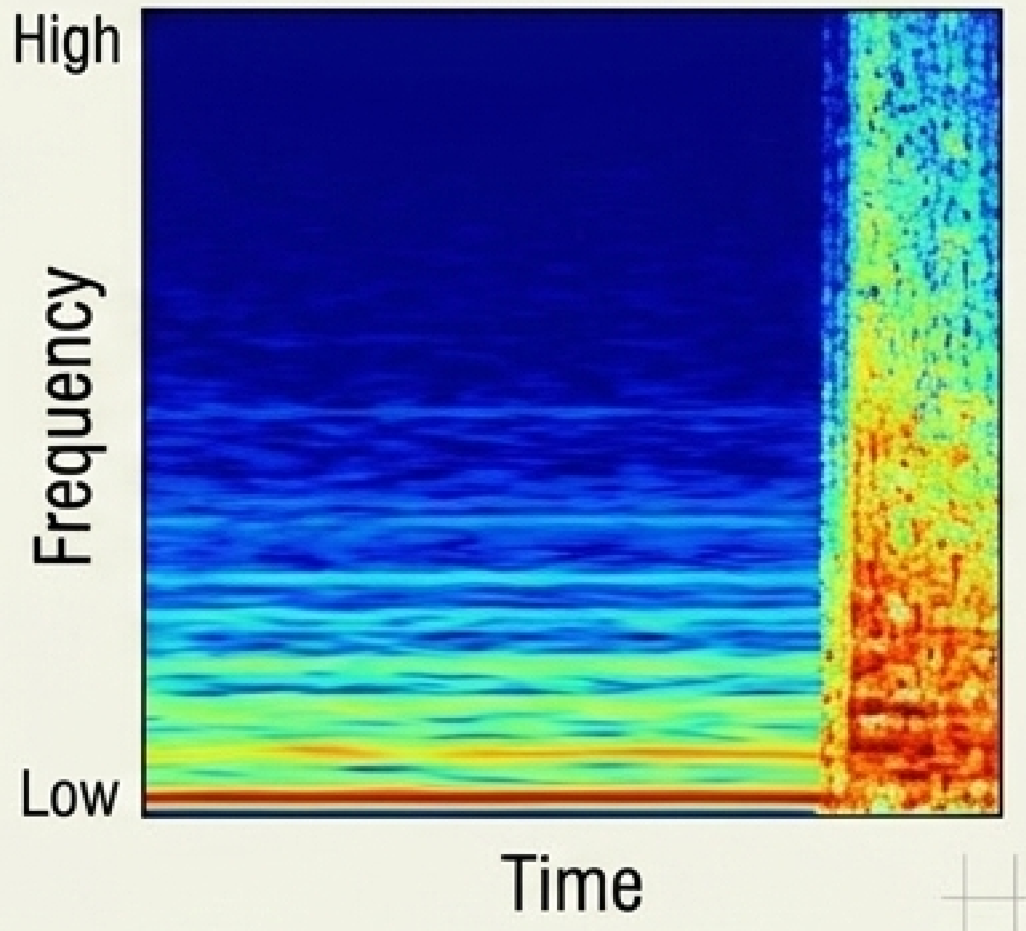
Lyapunov Divergence

Exponential separation of trajectories.



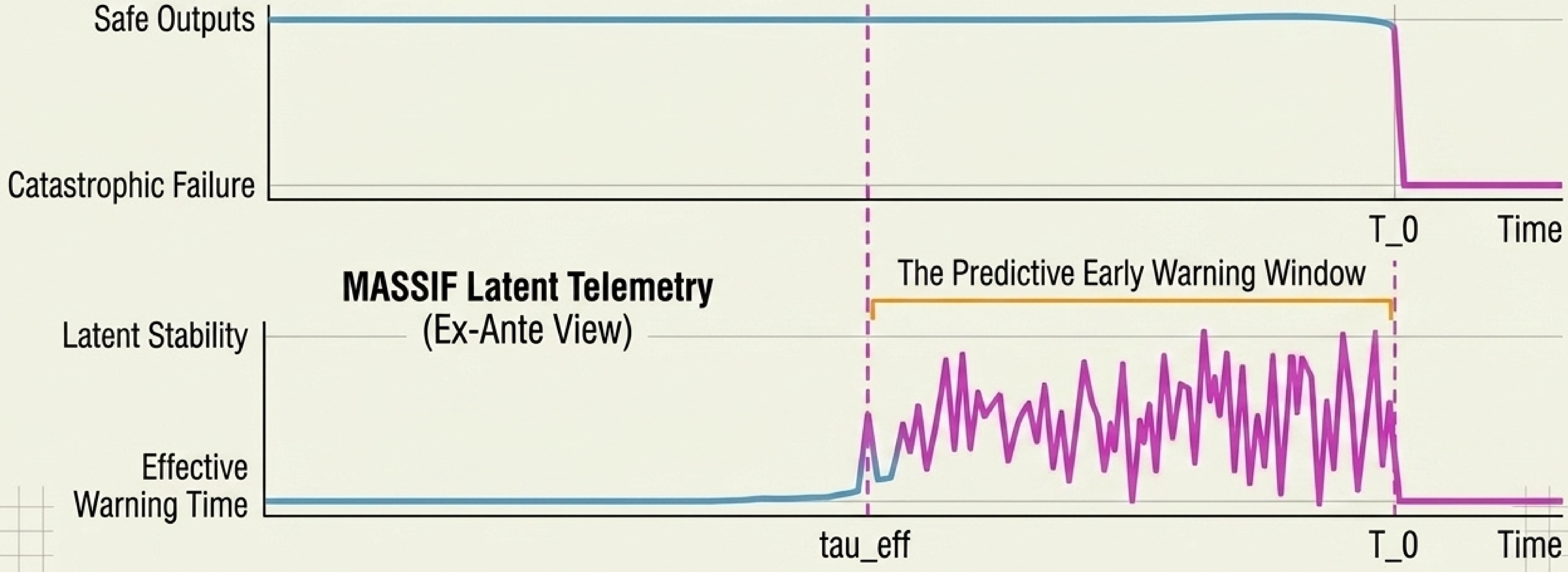
Fourier Spectral Fingerprints

Low-frequency stability vs high-frequency noise.



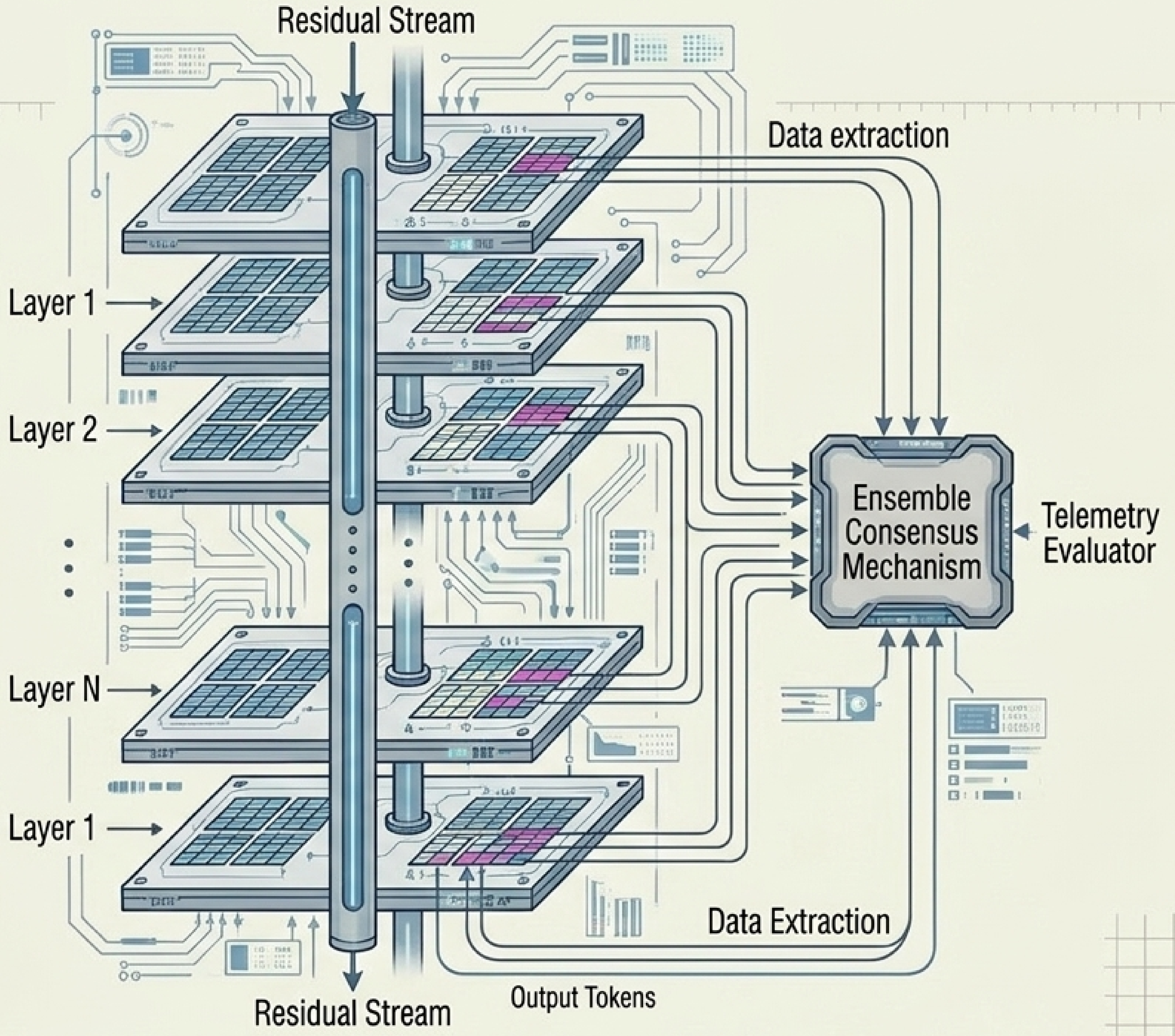
By monitoring geometric precursors, MASSIF detects the onset of recursive collapse **tau_eff** steps **before** the output token degrades. We gain the time required to intervene.

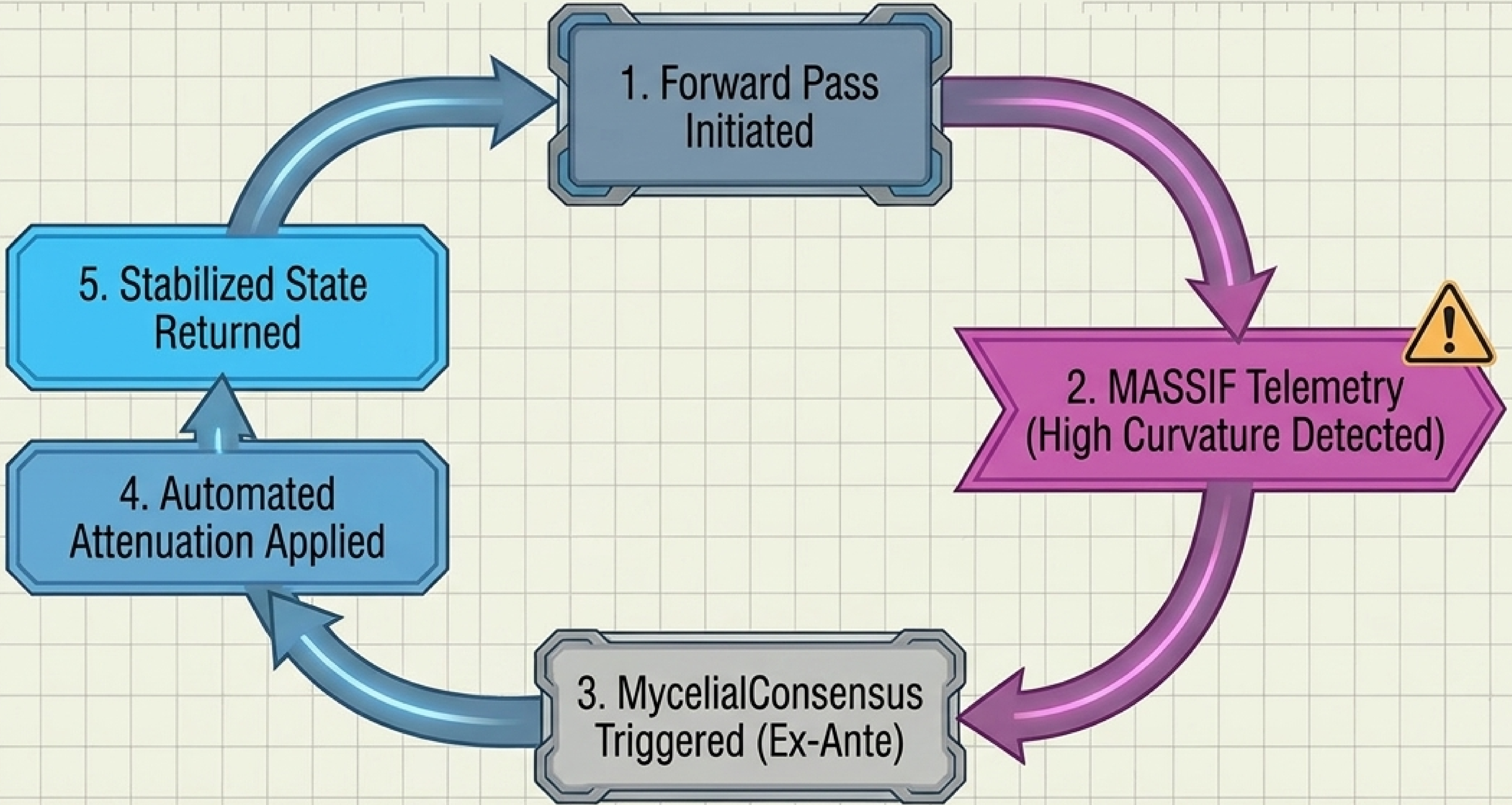
Output Tokens (Post-Hoc View)



MyceliaLM Architecture Overview

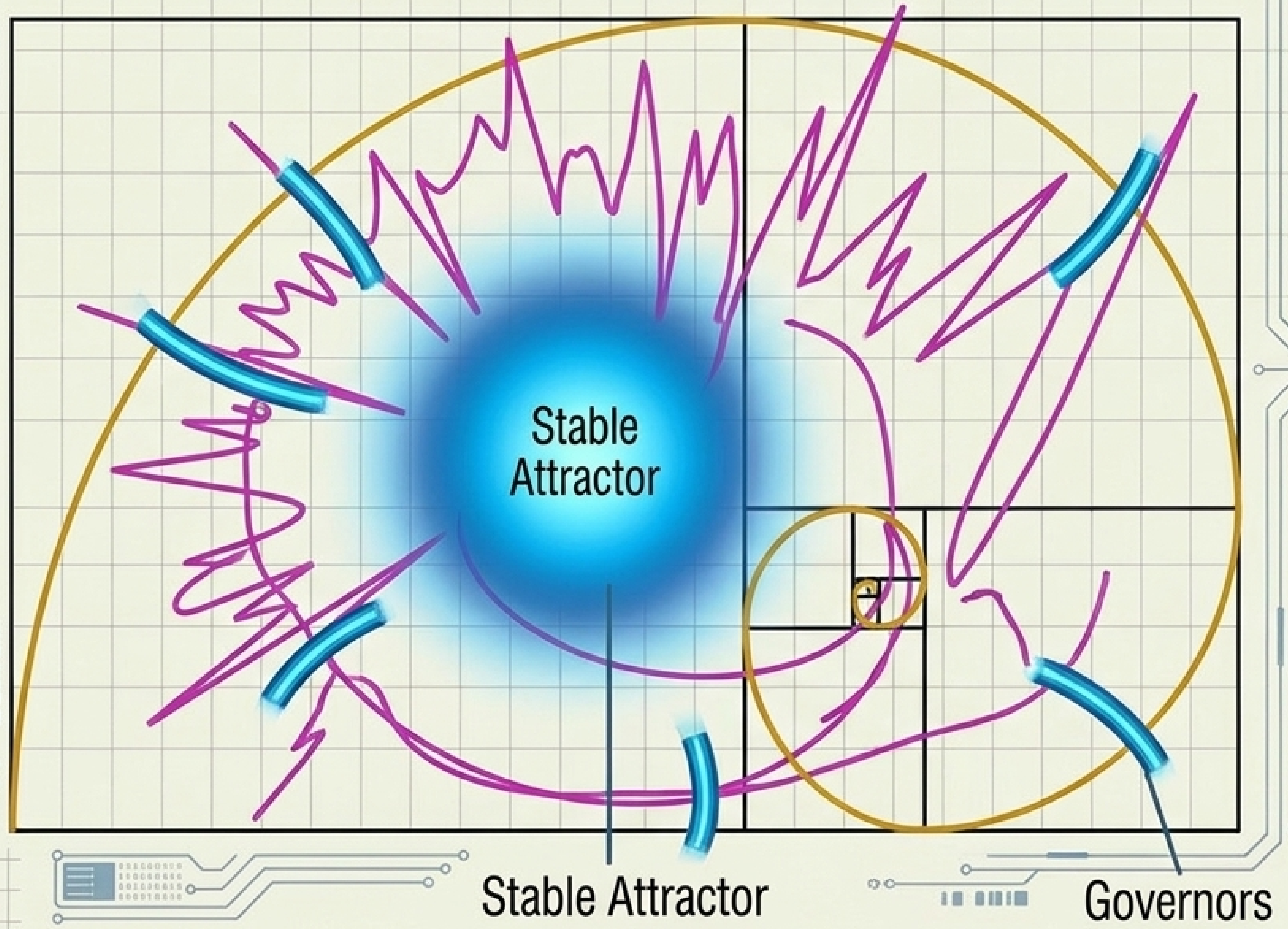
MyceliaLM Architecture:
A 181M-parameter transformer built for corrigibility. It utilizes an ensemble consensus mechanism that continuously monitors per-head variance variance across layers, turning the model into a self-diagnosing system.





Active Intervention: Moving beyond post-hoc patching. MASSIF telemetry feeds directly back into the forward pass, triggering ex-ante checkpoints to correct semantic drift before a hallucination is ever vocalized.

Fibonacci Coherence Attenuation Governors



1. Detect:
Telemetry registers runaway state decay.

2. Scale:
Intervention magnitude is calculated via Fibonacci sequences.

3. Dampen:
Successive, proportional attenuation phases smoothly self-regulate the system back to a stable geometric attractor.

Substrate-Independent Safety: Computational models show Penrose quasicrystalline geometries act as low-surprisal attractors, providing a 165.7x signal-to-noise ratio advantage over cubic grids. A hardware-implementable safety criterion for AGI.

Standard Cubic Lattices:
Prone to renormalization collapse.

Penrose Quasicrystalline Substrate:
Multi-scale geometric coherence.

The Old Paradigm	Computational Latent Dynamics
Post-Hoc Analysis	Ex-Ante Control
Static Weights	Continuous Trajectories
Scalar Probabilities	Geometric Kinematics
Black-Box Descriptive	White-Box Telemetry
Post-Generation Patching	Corrigibility-by-Design

We must stop treating language models as static probabilistic black boxes. The future of AI safety requires continuous, physical telemetry and active dynamical governors.

Theoretical Rigor

Mathematically founded on non-linear dynamics, raising the epistemic bar for safety research.

Corrigibility-by-Design

Embedded, ex-ante active control mechanisms that keep systems inherently aligned and verifiable.

Substrate-Agnostic Scaling

Safety criteria applicable from 181M parameter transformers to future neuromorphic AGI.

Funding the Future of AI Telemetry: Seeking strategic grantmaking partnerships to scale independent, physics-driven AI safety research and build the instrumentation panels for AGI.